

Class 8: Breast cancer mini project

Shuk Yee Leung (PID: A18050465)

Table of contents

Background	1
Data import	1
Clustering	3
Principal Component Analysis	4
The importance of data scaling	4
PCA of wisc.data	9
5. Combining methods	13
Clustering on PCA results	13
7. Prediction	16

Background

This mini-project explores unsupervised learning techniques applied to the Wisconsin Breast Cancer Diagnostic Data Set, which contains measurements of human breast mass cell nuclei. The project guides the user through exploratory data analysis, performing and interpreting Principal Component Analysis (PCA) to reduce the dimensionality of the data while retaining variance, and applying hierarchical clustering with different linkage methods. The ultimate goal is to combine PCA and clustering to better separate benign and malignant cell samples, evaluating the result using matrices like sensitivity and specificity, and finally demonstrating how to predict the classification of new samples using the developed PCA model.

Data import

Our data come from the U. of Wisconsin Medical Center

```
wisc.df <- read.csv("WisconsinCancer (3).csv", row.names = 1)
```

Q1. How many patients/samples are in this dataset?

```
nrow(wisc.df)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```
  B   M  
357 212
```

```
sum(wisc.df$diagnosis == "M")
```

```
[1] 212
```

Q3. How many variables/features in the data are suffixed with `_mean`?

```
colnames(wisc.df)
```

```
[1] "diagnosis"           "radius_mean"  
[3] "texture_mean"        "perimeter_mean"  
[5] "area_mean"           "smoothness_mean"  
[7] "compactness_mean"    "concavity_mean"  
[9] "concave.points_mean" "symmetry_mean"  
[11] "fractal_dimension_mean" "radius_se"  
[13] "texture_se"          "perimeter_se"  
[15] "area_se"             "smoothness_se"  
[17] "compactness_se"      "concavity_se"  
[19] "concave.points_se"   "symmetry_se"  
[21] "fractal_dimension_se" "radius_worst"  
[23] "texture_worst"       "perimeter_worst"  
[25] "area_worst"          "smoothness_worst"  
[27] "compactness_worst"   "concavity_worst"  
[29] "concave.points_worst" "symmetry_worst"  
[31] "fractal_dimension_worst"
```

```
length( grep("mean", colnames(wisc.df), value= T) )
```

```
[1] 10
```

There is a diagnosis column that is the clinical consensus that I want to exclude from any further analysis. We will come back later and compare our results to this diagnosis.

```
diagnosis <- as.factor(wisc.df$diagnosis)
head(diagnosis)
```

```
[1] M M M M M M
Levels: B M
```

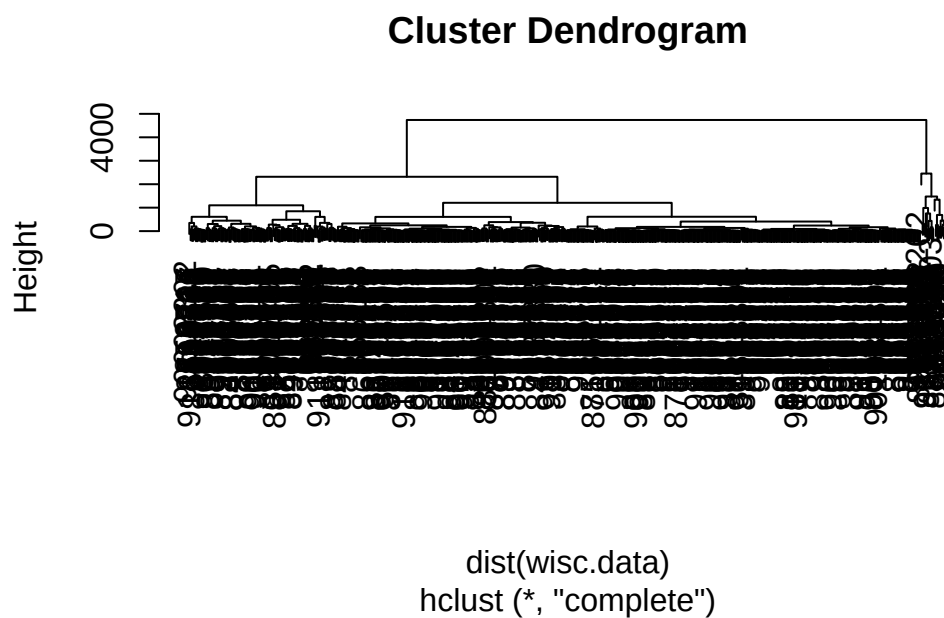
Now we can remove it from the wisc.df

```
wisc.data <- wisc.df[,-1]
```

Clustering

Let's try a hclus()

```
hc <- hclust( dist(wisc.data))
plot(hc)
```



We can extract clusters from this rather poor dendrogram/tree with the `cutree()`

```
grps <- cutree(hc, k=2)
```

How many individuals in each clusters?

```
table(grps)
```

```
grps
  1  2
549 20
```

```
table(diagnosis)
```

```
diagnosis
  B  M
357 212
```

We can generate a cross-table that compares our clusters `grps` vector with our `diagnosis` vector values

```
table(diagnosis, grps)
```

```
      grps
diagnosis  1  2
  B 357   0
  M 192  20
```

Principal Component Analysis

The importance of data scaling

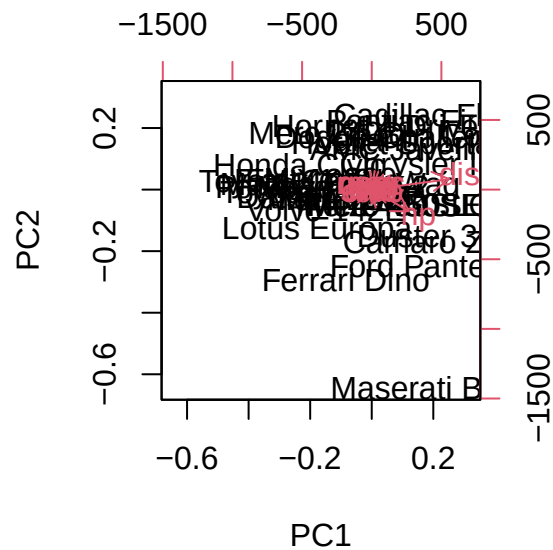
The main function for PCA in base R is `prcomp()`. It has a default input parameter of `scale=FALSE`.

```
#prcomp()
head(mtcars)
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

We could do a PCA of this data as is, and it could be mis-leading...

```
pc <- prcomp(mtcars)
biplot(pc)
```



Let's look at the mean values of each column and their standard deviation.

```
colMeans(mtcars)
```

mpg	cyl	disp	hp	drat	wt	qsec
20.090625	6.187500	230.721875	146.687500	3.596563	3.217250	17.848750
vs	am	gear	carb			
0.437500	0.406250	3.687500	2.812500			

```
apply(mtcars, 2, sd)
```

mpg	cyl	disp	hp	drat	wt
6.0269481	1.7859216	123.9386938	68.5628685	0.5346787	0.9784574
qsec	vs	am	gear	carb	
1.7869432	0.5040161	0.4989909	0.7378041	1.6152000	

We can “scale” this data before PCA to get a much better representation and analysis of all the columns

```
mtscale <- scale(mtcars)
```

```
round(colMeans(mtscale))
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	0	0	0	0	0	0	0	0	0	0

```
apply(mtscale, 2, sd)
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
1	1	1	1	1	1	1	1	1	1	1

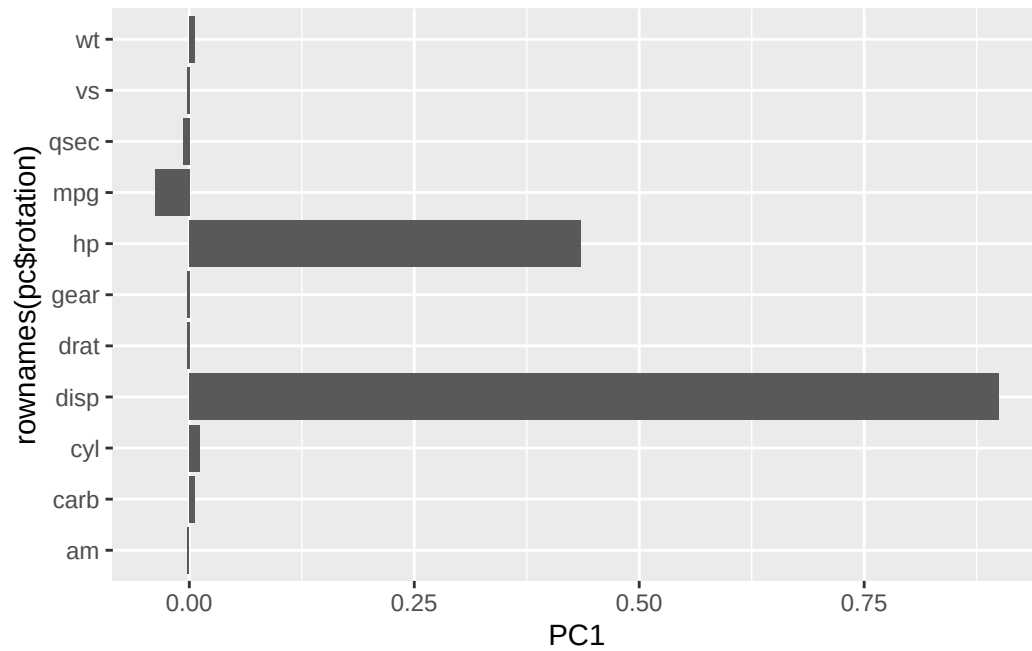
```
pc.scale <- prcomp(mtscale)
```

we can look at the two main results figures from PCA - the “PC plot” (a.k.a score plot, ordination plot, or PC1 vs PC2 plot). The “loadings plot” how the original variables contribute to the new PCs.

A loading plot of the unscaled PCA results

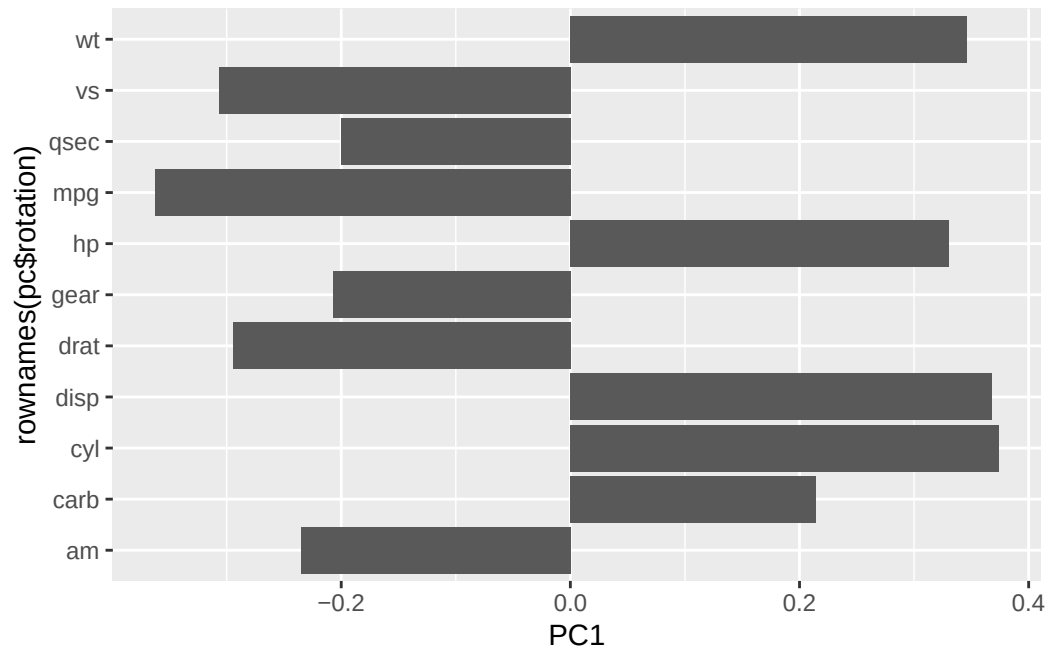
```
library(ggplot2)
```

```
ggplot(pc$rotation) +  
  aes(PC1, rownames(pc$rotation)) +  
  geom_col()
```



Loadings plot of the scaled data

```
ggplot(pc.scale$rotation) +  
  aes(PC1, rownames(pc$rotation)) +  
  geom_col()
```

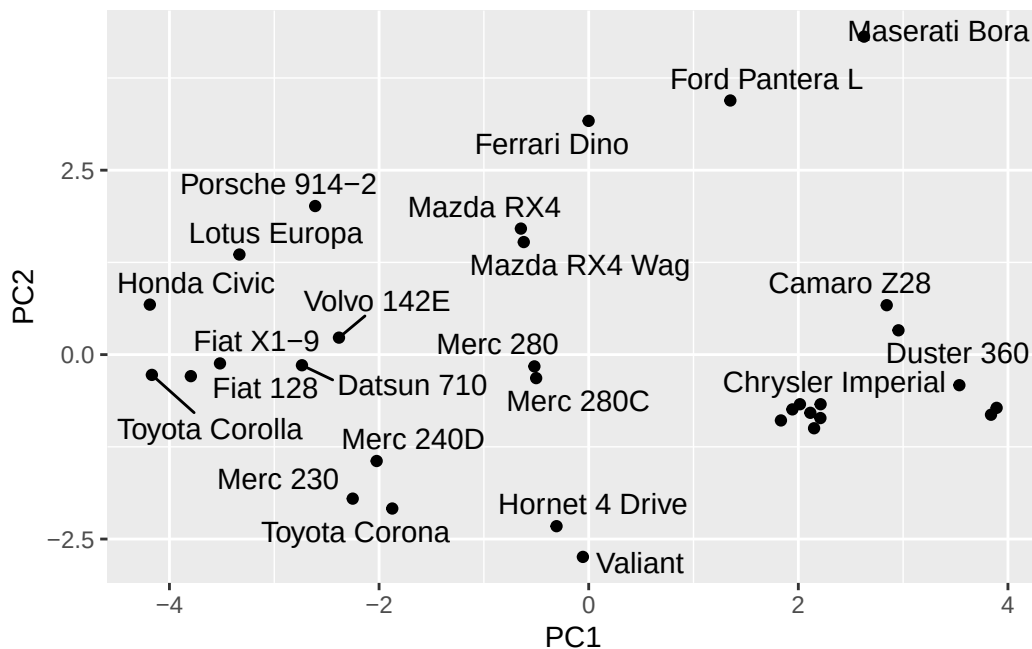


PC plot of scaled PCA results

```
library(ggrepel)

ggplot(pc.scale$x) +
  aes(PC1, PC2, label=rownames(pc.scale$x)) +
  geom_point() +
  geom_text_repel()
```

Warning: ggrepel: 9 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Key point: In general we will set `scale=TRUE` when we do PCA. This is not the default but probably should be...

We can check the SD and mean of the different columns in `wisc.data` to see if we need to scale - hiny: we do!

PCA of wisc.data

```
wisc.pr <- prcomp(wisc.data, scale=TRUE)
```

To see how well PCA is doing here in terms capturing the variance (or speed) in the data we can use the `summary()` function.

```
summary(wisc.pr)
```

Importance of components:

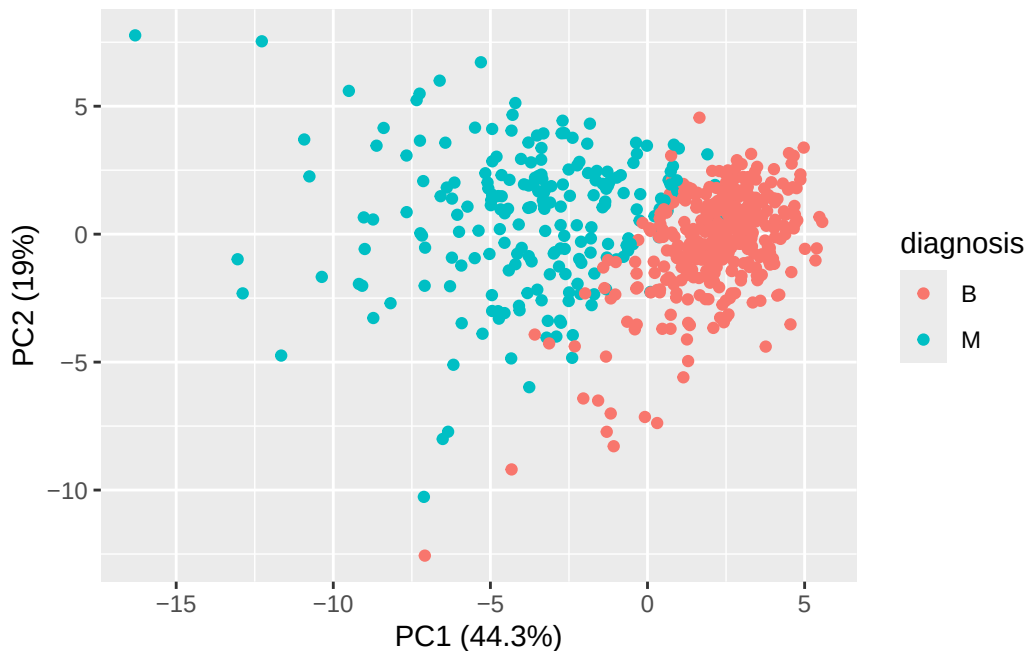
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010

	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.71852	0.65852	0.63171	0.61875	0.61875	0.61875	0.61875
Proportion of Variance	0.02251	0.02251	0.02251	0.02251	0.02251	0.02251	0.02251
Cumulative Proportion	0.93261	0.95492	0.97743	0.99994	1.00000	1.00000	1.00000

Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Let's make the main PC1 vs PC2

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point() +
  xlab("PC1 (44.3%)") +
  ylab("PC2 (19%)")
```



Q4. From your results, what proportion of the original variance is captured by the first principal components (PC1)?

44.3% (0.4427) of the original variance is captured by PC1.

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

3 PCs are required to describe at least 70% of the original variance.

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

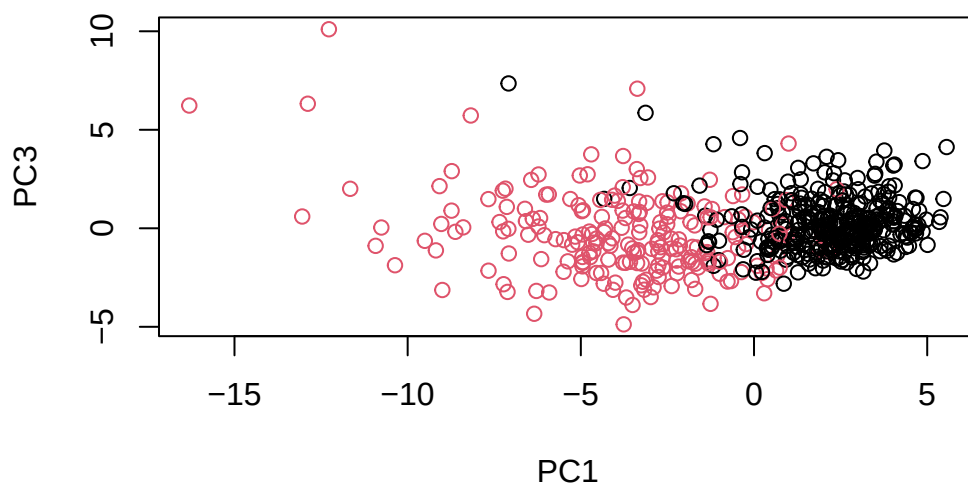
7 PCs are required to describe at least 90% of the original variance

Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

The points on the biplot were clustered together. There is 7 unlabeled data points because there were too many overlaps. It is difficult to understand because of too many data points clustered together and there is no clear separation of benign and malignant diagnoses groups.

Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
plot(wisc.pr$x[, c(1, 3)], col = diagnosis,  
     xlab = "PC1", ylab = "PC3")
```



Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`?

```
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

Q10. What is the minimum number of principal components required to explain 80% of the variance of the data?

```
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010

	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335

	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.39624	0.39624	0.39624	0.39624	0.39624	0.39624	0.39624
Proportion of Variance	0.00523	0.00523	0.00523	0.00523	0.00523	0.00523	0.00523
Cumulative Proportion	0.98335	0.98335	0.98335	0.98335	0.98335	0.98335	0.98335

Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

5 PCs required to explain 80% of the variance of the data

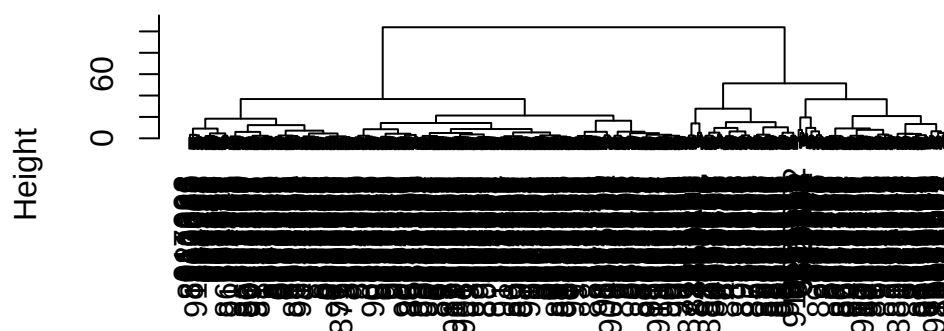
5. Combining methods

We can take our PCA results and use them as a basis set for other analysis such as clustering.

Clustering on PCA results

```
wisc.pr.hclust <- hclust( dist( wisc.pr$x[ ,1:2] ), method="ward.D2" )
plot(wisc.pr.hclust)
```

Cluster Dendrogram



```
dist(wisc.pr$x[, 1:2])
hclust (*, "ward.D2")
```

We can “cut” this tree to yeild our clusters (groups):

```
pc.grps <- cutree(wisc.pr.hclust, k=2)
table(pc.grps)
```

```
pc.grps
 1    2
195 374
```

How do my cluster grps compare to the expert diagnosis

```
table(diagnosis, pc.grps)
```

```
      pc.grps
diagnosis  1    2
B      18  339
M     177   35
```

```
table(diagnosis)
```

```
diagnosis
  B   M
357 212
```

Q12. Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.

Ward.D2 because it minimizes variance within clusters to make sure that the cluster are well-separated.

Q13. How well does the newly created model with four clusters separate out the two diagnoses?

```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)
# Compare to actual diagnoses
table(wisc.pr.hclust.clusters, diagnosis)
```

```
              diagnosis
wisc.pr.hclust.clusters  B   M
1      18 177
2     339   35
```

Q14. How well do the k-means and hierarchical clustering models you created in previous sections (i.e. before PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.km\$cluster and wisc.hclust.clusters) with the vector containing the actual diagnoses.

```
wisc.km <- kmeans(wisc.data, centers = 2)
table(wisc.km$cluster, diagnosis)
```

```
diagnosis
  B   M
1   1 130
2 356  82
```

```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=4)
table(wisc.pr.hclust.clusters, diagnosis)
```

```
              diagnosis
wisc.pr.hclust.clusters  B   M
1         0 112
2        18  65
3       232  18
4       107  17
```

They did really badly. we do much better after PCA - the new PCA variables (what we call a basis set) give as much better separation of M and B

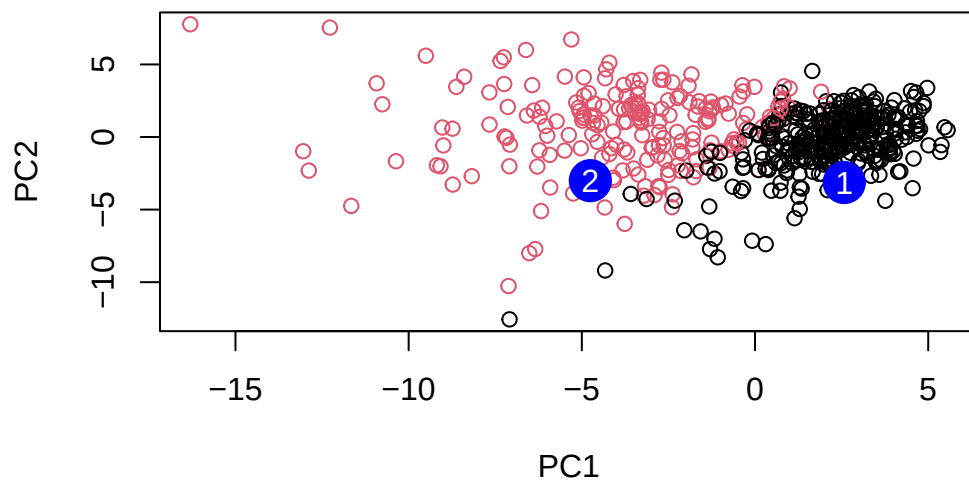
7. Prediction

We can use our PCA model for the analysis of new “unseen” data. In this case from U. Mich.

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
plot(wisc.pr$x[, c(1, 2)], col= diagnosis)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

People with malignant cancer (2).